Evaluating Neural Weather Forecasting Models During Extreme Weather Events

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Abstract

In recent years, deep learning has advanced weather forecasting by utilizing large-scale data and state-of-the-art physics-informed neural architectures. However, existing studies mostly neglect extreme weather event prediction, the most crucial application for mitigating environmental and economic impacts. In this study, we aim to evaluate and improve the performance of a neural ODE architecture in extreme weather conditions. Specifically, we train a physics-based model, ClimODE, on high-resolution weather data from the US and fine-tune it on a set of extreme weather events. Our experiments on the HR-Extreme dataset confirm that current deep learning systems, including those that model underlying physics, require better handling of extreme weather conditions. Further, fine-tuning demonstrates promising results for generalizing to extreme weather events. Our website is available at https://jenny-hrx.github.io/weather-site.

1 Introduction

Weather forecasting is a crucial part of human life; it influences everything from daily life to agricultural planning. Extreme weather events like storms, hurricanes, heatwaves, and cold spells have dangerous consequences on many aspects of our world. Mitigating the effects of these events is crucial to protect nature and people and minimize social and economic losses, making weather forecasting critical. Global climate change has led to increases in the frequency and severity of extreme weather events that are only expected to worsen in the future. Sequential extreme weather events have cascading effects that compound their consequences even further than those of just isolated events. In the face of climate change, it is more important than ever to develop accurate and timely weather forecasts that can accurately predict these events. However, as these events are exacerbated by climate change beyond our natural climate variability, they become even harder to predict.

Traditional climate and weather prediction models rely on complex numerical simulations of atmospheric physics. They are robust but have limitations, such as high computational costs and sensitivity to initial conditions. As deep learning (DL) and machine learning (ML) models have become more powerful, powerful DL models have been developed to analyze the weather. Deterministic models like ClimODE [18] and GraphCast [8] have achieved state-of-the-art (SOTA) performance in medium-range (10-day) global weather prediction that surpasses or matches traditional prediction models in accuracy while having computational costs that are several magnitudes lower. However, many such models are data-driven black-box models that neglect the underlying physical processes of the climate. Physics-based DL models have demonstrated potential in addressing these issues. ClimODE is one such model that learns weather evolution with physics-conserving dynamics. It is a spatiotemporal PDE climate and weather model with an ODE system tailored to numerical weather prediction.

Despite advances in ML models, predicting extreme weather events is still a critical and challenging issue in weather forecasting. AI methods produce smoother forecasts, which risk underestimating the magnitude of extreme weather events. Popular weather datasets, such as ERA5, used to train these models have very few examples of such events due to their rarity, so models cannot be trained or tested on them. As a result, ML models struggle to generalize to extreme weather events due to their limited representation in available data.

In this study, we explore how the performance of deterministic ML models on extreme weather events can be improved. To answer this, we propose first investigating the question: how do current deterministic ML models generalize to extreme weather events? Then, we will examine how fine-tuning with extreme weather data can improve generalization.

Testing existing models on extreme weather datasets is a crucial first step in understanding and addressing their limitations in generalization. Data augmentation with extreme samples can be used for models to learn these events and improve their generalization, and it is a promising method to address this challenge. New datasets focused on extreme weather events like HR-Extreme [12] have recently been developed, making it possible to start improving model performance.

Physics-guided AI approaches can increase model consistency with the physical world, especially on unseen or extreme events. Models such as ClimODE have made significant progress in the integration of physics and AI, but further work is needed for optimization. We adapt and train a regional ClimODE model to use as our base model and evaluate its capabilities in predicting extreme weather events. Our results show weaker performance of ClimODE on extreme events, which we successfully improve with fine-tuning.

2 Related Work

Historically, physics-based models, such as Numerical Weather Prediction (NWP) models [5], have been the gold standard in weather forecasting. The Integrated Forecast System (IFS) [6] is considered to be the world's best NWP. These models solve large-scale Partial Differential Equations (PDEs) that represent the mathematical equations that govern atmospheric, oceanic, and terrestrial systems. While they are successful, they have several significant limitations [16]. Firstly, they are very computationally intensive due to the large amount of data preprocessing and complexity of the governing PDEs. Additionally, they are limited in capturing the

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intricacies of weather processes; the underlying equations rely on simplifications about the atmosphere. Finally, they exhibit sensitivity to initial conditions, structural discrepancies across models, and regional variability [18].

Recently, DL models have emerged as a strong alternative to NWPs. They analyze complex climate data by learning intricate dependencies and can provide quick predictions after being trained. Various ML techniques have outperformed the IFS [1], particularly when leveraging a multitude of variables and exploiting correlations among them.

2.1 Deterministic Predictive Models

Predictive learning methods are usually deterministic: models predict future values of weather variables based on past and present observations. These models learn weather patterns and dependencies by minimizing a point-wise loss function (mean absolute error). These deterministic predictive models can be further split into general-purpose large models and domain-specific models [16]. General-purpose models are usually trained on large, global datasets with multiple variables of interest. This allows them to perform global weather prediction for a broad range of applications. On the other hand, domain-specific models predict a single variable and are applied to regional weather prediction. Domain-specific models are typically trained for fixed datasets and struggle to generalize beyond their training conditions [19]. In comparison, general-purpose models are more adaptable to task-specific applications; they have extensive pretraining and can adjust to new conditions with limited training.

2.1.1 General-Purpose Predictive Models. Our work will focus on general-purpose models. Models within this category usually fall into three types [16]: GNN-based models, transformer-based models, and physics-AI-based models.

GNN-based models: Graph neural networks (GNNs) are used to represent the Earth as a graph [7]. By using nodes to represent spatial locations and edges to encode the relationships between them, GNNs capture spatial dependencies in weather both globally and locally. GraphCast [8] forecasts hundreds of weather variables at a high spatial resolution of 0.25 degrees with a forecast range of up to 10 days. It has stronger performance than the European Centre for Medium-Range Weather Forecasts (ECMWF)'s High-RESolution forecast (HRES), which is part of the IFS. GraphCast's performance indicates that ML models can compete with traditional weather models.

Transformer-based models: While GNN-based models excel at modelling spatial relations, transformer-based models are suitable for learning temporal relations. Many models widely use transformers as a backbone. Although they have impressive performance, iterative inference processes accumulate errors as the prediction window increases. To address this, Pangu-Weather [2] implements a hierarchical temporal aggregation strategy for medium-range forecasting. Pangu-Weather was trained on 39 years of global weather data. It performed better than IFS on deterministic forecasting on reanalysis data from ERA5 and was 10,000 times faster.

Physics-AI-based models: Although data-driven models like Pangu-Weather have strong performances, they are black-box models that neglect the underlying physics of weather, like convection, turbulence, and airflow. They also lack uncertainty quantification for their predictions. Physics-AI-based models show potential in alleviating these issues. ClimODE uses advection to develop a spatiotemporal climate and weather model. It models weather changes due to spatial movement over time with value-conserving dynamics with a continuous-time PDE. It uses a Gaussian emission network to estimate the uncertainty quantification. Conformer [14] is a spatiotemporal Continuous Vision Transformer model that implements continuity in the multi-head attention mechanism to learn the continuous weather evolution over time.

2.2 Extreme-Weather Generalization

Accurately predicting extreme weather events, such as tornadoes, storms, and hurricanes, is crucial for enabling effective disaster preparation and response. However, current datasets and models are limited in this regard.

2.2.1 Datasets. Popular existing weather benchmarks are typically focused on general weather prediction. Extreme climate events are driven by complex interactions that are not seen in these simplified and controlled benchmarks. WeatherBench is a popular general weather dataset derived from ERA5 data [13]. It is used for training and evaluating weather-forecasting models, but it omits extreme events. The goal of the benchmark is to provide a simple and clear problem, and users are encouraged to verify extreme weather events separately. WeatherBench does not enable model finetuning or transfer to other tasks. Since a goal of such general weather models is to be an efficient backbone for a range of tasks, they need to be tested more comprehensively. In the context of rapid climate change, evaluating model adaptation to extreme climate events is crucial to ensure that they can be used in the decision-making process.

Datasets developed to address extreme weather lack a variety of extreme weather types, have low-quality data, or use data that is incompatible with weather forecasting models. The ExtremeWeather dataset [11] is limited to three specific extreme weather events. EWELD [9] is a weather dataset that uses multi-channel satellite data, ground observation data, and soundings for various types of extreme weather in China. However, its feature maps have low resolution, contain a limited number of extreme weather types, redundant data, and less accurate location information.

To address these gaps in extreme weather evaluation, researchers have recently developed new datasets. HR-Extreme [12] provides a real-time dataset comprising high-resolution extreme weather cases in the U.S. It consists of a comprehensive set of 17 extreme weather types, like hail, tornadoes, and heavy rains. The data is from the High-Resolution Rapid Refresh (HRRR) provided by the National Oceanic and Atmospheric Administration (NOAA). The 3-km resolution is significantly higher than the roughly 33-km resolution of the WeatherBench dataset.

2.2.2 ML Models. Current SOTA deep learning models like Pangu-Weather [2], and Fuxi-Extreme [20] for weather forecasting either lack testing in extreme weather or have only been tested on a limited range of extreme weather events. Fuxi-Extreme uses a denoising diffusion probabilistic model to restore finer-scale details in the surface forecast data generated by the SOTA FuXi model [4] in 5-day forecasts. However, they only assess performance on tropical cyclone forecasting. To improve extreme-weather generalization, [10] trains deep learning models to forecast extreme heat events. Their results showed that models trained with their architecture on MSE loss had poor forecasts of extremely hot days. They found that training models with custom loss functions that prioritize extreme values significantly improved heatwave prediction while having negligible skill loss in general temperature prediction, showing that models trained to predict extremes still retain skill in general settings. Their work highlights the weaknesses of current models in forecasting extreme events, but is limited to extreme heat. Our work further explores the performance of a SOTA model on a more comprehensive set of extreme weather events and the use of this extreme data for fine-tuning.

3 Methods

SOTA neural weather models are often superior to NWP in terms of requiring orders of magnitude less computation during inference and capturing non-linear relationships and spatiotemporal dynamics of Earth's weather properties [8, 18]. However, the robustness of these general-purpose forecasting models when predicting extreme weather events remains understudied, despite the critical importance of reliable predictions for disaster preparedness and public safety [12, 19]. It remains uncertain whether these models can reliably forecast weather patterns amidst the profound and unpredictable climate changes anticipated in the coming decades.

3.1 Models

Our primary focus is on evaluating and improving ClimODE, which is more computationally efficient than other SOTA models and can be trained on a single GPU [18]. Models like Pangu-Weather [2] and GraphCast [8] leverage large-scale, long-term weather data and utilize tens to hundreds of GPUs during training; they also employ SOTA architectures to forecast without explicitly embedding physics. Physics-guided AI approaches like ClimODE can make models more consistent with the physical world, especially on unseen or extreme events, which may be an advantage for this task. Thus, we use ClimODE as the base model for our experiments.

3.1.1 *Problem Formulation.*

Weather state is represented as a spatiotemporal feature map, $\mathbf{U} \in \mathbb{R}^{K \times T \times H \times W}$, defined over a latitude-longitude grid. K is the number of weather variables, T is the number of time steps, H and W are latitude and longitude, respectively.

Weather forecasting involves predicting the weather state $\mathbf{U}_t$ at time t given historical observations $\mathbf{U}_{t^-} = \{\mathbf{U}_{t-1}, \mathbf{U}_{t-2}, \dots\}$:

$$\mathbf{U}_t = f_\theta(\mathbf{U}_{t^-}) \tag{1}$$

where f_θ is a learned neural network parameterized by θ. This general framework can be instantiated with many different architectures, including but not limited to autoregressive transformers, GNNs, and neural ODEs [2, 8, 18].

3.1.2 *ClimODE.*

ClimODE is a physics-informed neural ODE approach that is far more efficient than completely data-driven DL approaches such as GraphCast and Pangu-Weather [18]. Rather than learning atmospheric dynamics purely from data, it embeds an *advection equation* directly into the architecture:

$$\frac{\partial \mathbf{U}_t^{(k)}}{\partial t} = -\mathbf{V}_t^{(k)} \cdot \nabla \mathbf{U}_t^{(k)} - \mathbf{U}_t^{(k)} \nabla \cdot \mathbf{V}_t^{(k)} \tag{2}$$

where $\mathbf{V}_t^{(k)}$ is the velocity field of the weather state k at time t. The first term in Eq. 2 denotes the advective transport, and the second term accounts for the compression/expansion effects.

A learned velocity field dictates the spatiotemporal movement of weather quantities, through transport and compression terms, with the advection formulation ensuring mass conservation [18]. The transport term represents the amount of movement for each weather quantity over a specific time and location, while the compression term determines whether the weather quantity compresses, spreads out, or passes through certain points. Velocity is modeled using an NN that takes the current weather state, its gradient, current velocity, and spatiotemporal embeddings as input, and outputs the change in velocity. This second-order approach improves performance compared to simply predicting the velocity [18]. The NN architecture comprises a convolutional module that captures local dependencies and an attention module that can capture global relationships. Finally, to account for value gain or loss and uncertainty, they utilize a Gaussian emission network that outputs the mean and variance, representing the value gain and uncertainty in the prediction, respectively. Due to its physics-informed neural ODE architecture, ClimODE can be trained efficiently on a single GPU.

3.2 Datasets & Preprocessing

We used two datasets covering weather data from the US in 2020. We leveraged the HRRR data source to sample a dataset to use for training ClimODE. For evaluation, we utilized the HR-Extreme dataset, which comprises various extreme weather events from 2020 extracted from HRRR.

3.2.1 *High-Resolution Rapid Refresh (HRRR).*

NOAA's High Resolution Rapid Refresh (HRRR) dataset provides hourly atmospheric data at 3km resolution covering the US in a 1799×1059 pixel grid (spanning $21.1^\circ - 52.6^\circ$ latitude and $225.9^\circ - 299.1^\circ$ longitude). HRRR is better suited than existing benchmarks, such as ERA5, in terms of high resolution and time granularity, which enables the capture of high-quality data for extreme events.

Since HR-Extreme's date range is from July 2020 to December 2020, we extract data prior to July 2020 for use in training. We randomly sampled various crops from the 1799×1059 grid and time intervals. This allows us to have a representative training set of various regions and time intervals in US. Each data point represents a 3-hour interval of crops of shape 320×320, matching the format in HR-Extreme. We normalize each variable based on the minimum and maximum values in the training set. We utilized the Herbie library for constructing this dataset [3].

3.2.2 *HR-Extreme.*

HR-Extreme is a recent weather benchmark covering 17 types of extreme events that occurred in 2020. Some of the extreme events include debris flow, flood, and extreme heat [12]. HR-Extreme utilizes a much higher resolution, 3x3 km, which is ten times higher than the standard benchmark ERA5, with a resolution of $0.25^\circ \times 0.25^\circ$. The data from the hour of the extreme event and the two hours before are included in the dataset. There are 69 weather-related variables associated with each grid point and timestep. We extract the five relevant weather variables used in ClimODE for prediction. This step greatly reduced the computational and storage costs. We use most of HR-Extreme as an evaluation set while leveraging a subset as training and validation sets for fine-tuning.

Debris Flow	Marine Strong Wind
Flash Flood	Marine Thunderstorm Wind
Flood	Thunderstorm Wind
Funnel Cloud	Tornado
Hail	Waterspout
Heavy Rain	Wind
Lightning	Heat
Marine Hail	Cold
Marine High Wind	

Table 1: HR-Extreme event types [12].

3.2.3 HR-Extreme Data Extraction. HR-Extreme derives its data from HRRR. The authors identify extreme weather events using three sources: the NOAA Storm Events Database, the NOAA Storm Prediction Center's daily reports of hail/tornadoes/wind (which they cluster using DBSCAN to identify event boundaries from scattered user reports), and manually filtered extreme temperature cases (using 0.05 and 0.95 percentile thresholds of $-29C^\circ$ and $37C^\circ$).

For each identified event, they extract the relevant region from the full HRRR grid using the event's timestamp and area, then cover the event's area with a 320x320 grid, resulting in a 960x960 km region. They provide binary masks to indicate which pixels fall within the event area.

The final dataset contains thousands of extreme weather events recorded hourly for the year 2020. The data from January 2020 to June 2020 is used for training, while the data from July 2020 to December 2020 is used for testing. Each event is annotated with a specific event type, time, and location in the grid.

Although the dataset has 69 weather variables at different pressure levels, we extract five of them, which are the weather variables that ClimODE predicts [18]: 2 meter temperature (t2m), 10 meter U wind component (u10), 10 meter V wind component (v10), Atmospheric Geopotential (z), and Atmospheric Temperature (t).

Table 3.2 shows the key characteristics of the constructed datasets, HRRR and HR-Extreme.

3.3 Retraining ClimODE

The weights for ClimODE's regional North America model were unavailable, so we had to retrain ClimODE. Although we could have retrained the regional ClimODE model using the same WeatherBench data that the authors originally used, we chose to retrain it using HRRR data.

While high-resolution weather datasets like HR-Extreme are able to capture fine-scale atmospheric features and extreme weather events, they are not as compatible with lower-resolution models like ClimODE. ClimODE was trained on ERA5 data from WeatherBench that has 5.625° resolution, which results in 32×64 grids, lacking the necessary resolution to capture local extreme events. Retraining with HRRR data avoids the mismatch between training and evaluation resolutions to provide a better idea of ClimODE's extreme-weather forecasting abilities; it may not have been able to perform as well at forecasting high-resolution events if it were trained at a much lower resolution. The low-resolution ERA5 data originally used to train ClimODE makes it less effective for predicting more localized weather events like severe thunderstorms. Extreme events with limited spatial range are more clearly represented in HRRR data but may be absent or vague in ERA5.

Additionally, the HRRR data goes up to May 2025, which is more recent compared to ERA5, which does not go past 2018. This recency will provide better training data due to the increased extremity of weather over time. Using more recent weather data to retrain could provide a better picture of the ability of current ML models to accurately predict more extreme weather events.

3.3.1 Modifications to ClimODE Framework. While we mostly preserved the original ClimODE architecture, some modifications were necessary in order to adapt it to HR-Extreme. Firstly, since HR-Extreme consists of independent 3-hour windows rather than a long sequence of contiguous weather observations. Therefore, we modified the implementation to be able to feed these 3-hour windows into the model, which was originally designed to process the whole sequence as a single point. This modification extended ClimODE to handle missing data and efficiently process high-resolution data in batches. Secondly, ClimODE requires initial velocities to start the ODE solving process and estimates the initial velocities using the past three timesteps. While this is trivial to apply to a continuous time series formatted ERA5 dataset, it is not possible in HR-Extreme since we have 3-hour windows. Therefore, we used the first two time steps to estimate the initial velocity. In addition, they used a kernel-based smoothing in order to obtain spatially smooth velocities. However, the kernel K they compute has dimensions $(H * W, H * W)$ that handles the distance between every possible pair of grid points, where H and W are grid dimensions. While this is easy to compute and store for global 32×64 grids, it is infeasible for high-resolution 320×320 crops, which require a kernel of billions of entries. Therefore, we replace the kernel smoothing loss with a local smoothing loss that computes the velocity difference between adjacent pixels in 320×320 crops. Thus, the initial velocity estimation problem is formulated as:

$$\hat{v}_k(t) = \arg\min_{v_k(t)} \left\| \tilde{\dot{u}}_k(t) + v_k(t) \cdot \tilde{\nabla} u_k(t) + u_k(t) \tilde{\nabla} \cdot v_k(x,t) \right\|_2^2 + \alpha \left\{ \cancel{\|v_k(t)\|_K^2}\ \|\nabla v_k(t)\|_2^2 \right\}$$

where $\|\nabla v_k(t)\|_2^2$ denotes the norm of the spatial gradient of the estimated velocities, penalizing rapid changes to enforce local smoothness.

4 Experiments

To answer our question of how the performance of ML models on extreme weather events can be improved, the experiments that we will conduct are two-stage:

First, we evaluate the effectiveness of ClimODE on extreme weather events to answer our question of how deterministic SOTA ML models generalize to extreme weather events. We evaluate

Table 2: Performance of ClimODE trained on HRRR in HRRR test set and HR-Extreme.

Variable	Metric	HRRR	HR-Extreme	HR-Extreme (Fine-tuned)
t2m	RMSE	17.01	14.85	11.52
	ACC	0.43	0.43	0.47
u10	RMSE	11.79	11.77	7.78
	ACC	0.22	0.35	0.50
v10	RMSE	8.66	8.88	5.93
	ACC	0.26	0.41	0.51
z	RMSE	130.98	159.36	121.30
	ACC	0.46	0.27	0.28
t	RMSE	12.66	14.36	9.98
	ACC	0.50	0.24	0.31

Table 3: Characteristics of HRRR dataset constructed for training and HR-Extreme benchmark.

	HRRR	HR-Extreme
Windows	8,372	16,302
Period	Jan-Jun 2020	Jul-Dec 2020
Size	31 GB	48 GB
Interval	1 h	1 h
Window size	3	3
Resolution (px)	320x320	320x320
Resolution (km)	3x3 km	3x3 km

ClimODE's predictions of the key meteorological variables tested in the original ClimODE paper: atmospheric temperature (t), ground temperature (t2m), geopotential (z), and ground wind vector (u10, v10) using the HR-Extreme dataset.

Then, we test if fine-tuning improves extreme-weather performance. We fine-tune ClimODE on a subset of 30% of HR-Extreme and test it on the remaining 70% of HR-Extreme. We then analyze the change in performance compared to the original general model.

4.1 Training Setup

4.1.1 HRRR. First step of our experiments was to train ClimODE on HRRR dataset we extracted. We trained the model on 3 NVIDIA A40 48GB GPUs for 30 epochs, which took around 15 hours. We saved the model checkpoint with the lowest validation loss. We split the dataset into 80% training, 10% validation and 10% test sets without any sort of shuffling that would result in data leakage. We used a learning rate of 5×10^{-4} with cosine annealing scheduler and weight decay rate of 1×10^{-3}, following the original implementation. Finally, we used a batch size of 4.

4.1.2 HR-Extreme. To fine-tune ClimODE, we split the HR-Extreme dataset into 20% train, 10% validation, and 70% test sets. We used a learning rate of 1×10^{-4} and a batch size of 2. Similarly, we saved the model weights with the lowest validation loss across 10 epochs.

4.2 Evaluation Metrics

There are two major evaluation metrics used in weather forecasting: Root Mean Squared Error (RMSE) and Anomaly Correlation Coefficient (ACC)[2, 8, 12, 18, 19]. These metrics usually factor in the Earth's curvature using weightage by latitude to account for the difference in the covered area between grid points [2, 18]. RMSE assesses model accuracy in capturing spatial or climate patterns by measuring absolute prediction errors, and ACC measures a model's ability to predict deviations from normal conditions. Lower RMSE values and higher ACC values indicate better model performance.

The equations are as follows:

$$\text{RMSE} = \frac{1}{N}\sum_{t=1}^{N}\sqrt{\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\alpha(h)\left(y_{thw}-u_{thw}\right)^2} \tag{3}$$

$$\text{ACC} = \frac{\sum_{t,h,w}\alpha(h)\tilde{y}_{thw}\tilde{u}_{thw}}{\sqrt{\sum_{t,h,w}\alpha(h)\tilde{y}^2_{thw}}\sqrt{\sum_{t,h,w}\alpha(h)\tilde{u}^2_{thw}}} \tag{4}$$

where $\alpha(h) = cos(h)/(1/H)\sum_{h'}^{H} cos(h')$ is the latitude weight and $\tilde{y}$ = y - C and $\tilde{u}$ = u - C are averaged against empirical mean C $=1/N\sum_t y_{thw}$.

4.3 Results

4.3.1 How does ClimODE generalize to extreme weather events? Table 2 shows the performance of ClimODE on the HRRR test set and HR-Extreme. Although the performance in both datasets is not competitive due to the small training set covering a short period of time, it can be greatly enhanced if the model is trained for a longer period with large-scale, long-term data. Figure 2 shows that the model consistently reduces the training loss while the validation loss saturates after $15-20$ epochs, indicating that the model can learn given enough training set.

Regarding the performance discrepancies between the normal weather data in HRRR and HR-Extreme, which represents extreme weather conditions, the performance of two out of five weather variables is significantly lower on HR-Extreme. This indicates that the model trained on randomly sampled weather data underperforms in extreme weather conditions, suggesting that special treatment is necessary for ML models to be used for extreme weather prediction.

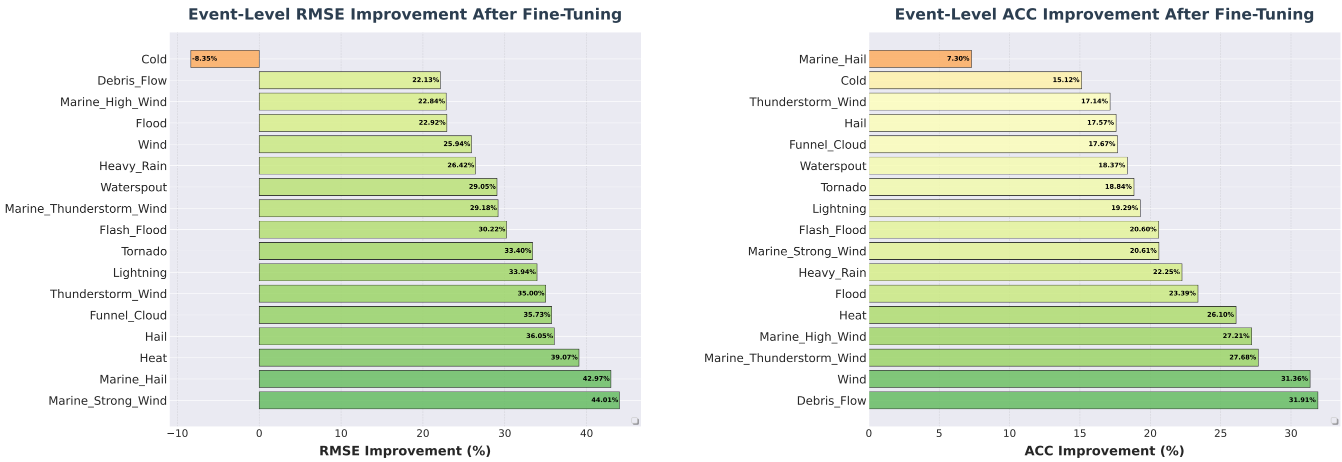

Figure 1: Performance improvements after fine-tuning across extreme weather event types

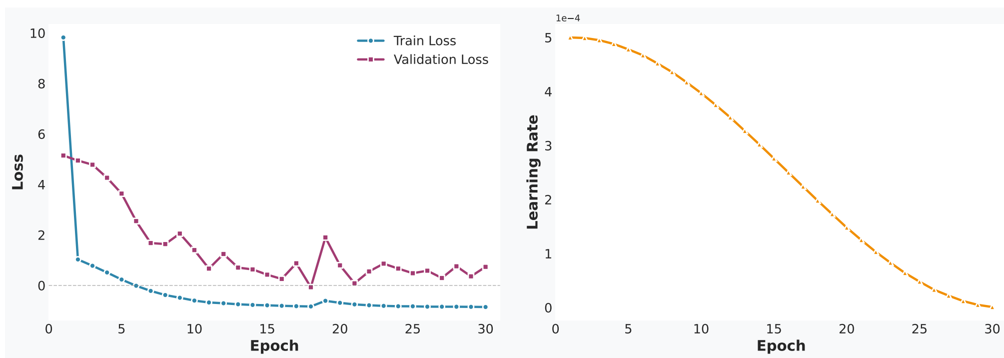

Figure 2: The evolution of loss and learning rate across epochs.

4.3.2 Can fine-tuning improve extreme-weather performance? In the second stage, we aim to fine-tune ClimODE on HR-Extreme to better capture the extreme weather conditions. The fine-tuned model had improved performance on HR-Extreme compared to the regular model's performance on both HR-Extreme and HRRR in all metrics except for ACC for z and t. Although the current metrics are not very strong, the improved performance still clearly shows the potential of using fine-tuning to improve the generalization of ML models to extreme-weather events. With a stronger base model trained on more data, ClimODE would have much more competitive performance. Further, Figure 1 shows the event-level performance improvement after fine-tuning. Fine-tuning achieves substantial performance improvements across nearly all extreme weather event types with RMSE reductions of 22-44% and ACC gains of 15-32%, though Cold events uniquely show RMSE degradation, likely because they represent a different distribution from other event types. Specifically, the temperature (t2m and t) and geopotential (z) exhibit different behavior for extreme cold events compared to other event types. Refer to Appendix A for more detailed results and distribution plots.

The z and t variables had the largest range of possible values among the variables, which is likely why the model struggled the most on their ACC. Standard loss functions can produce overly smooth forecasts that avoid large deviations. Although the RMSE and ACC improved for z and t after fine-tuning, their ACC did not outperform the standard model on HRRR, unlike the other variables, which suggests that the model likely learned to produce a smooth prediction for z and t that improved RMSE at the cost of ACC. Fuxi-Extreme showed that models trained with their architecture on MSE loss had poor forecasts of extremely hot days. They found that training models with custom loss functions that prioritize extreme values significantly improved heatwave prediction while having negligible skill loss in general temperature prediction. Although fine-tuning did improve ACC for z and t, the improvement was likely limited by the loss function; using a more trained base model with a custom loss function prioritizing extremes is probably needed to maximize performance. Combining physics-based models like ClimODE with custom loss functions would likely improve extreme-weather generalization even further and would be an interesting avenue for future work.

5 Conclusion & Future Work

In this study, we worked on assessing and improving the forecasting performance of ClimODE on extreme weather events observed in the US over a short period of time. To do so, we made two changes to the original ClimODE framework: processing independent 3-hour windows and using local smoothing loss instead of kernel-based smoothing for initial velocity estimation. We observed a performance discrepancy between the model's performance on normal and extreme weather conditions and successfully used fine-tuning to reduce this discrepancy.

As discussed in the methods section, given more time, the base model would have ideally been retrained with a larger high-resolution grid covering the entire US over a continuous period of time. This would be the most immediate next step of future work. Although our ClimODE model provides an idea of extreme weather performance, to evaluate its full capabilities, it would need to be retrained in this manner. Comparing physics-based models like ClimODE to GNN and transformer-based models like GraphCast and Pangu-Weather to evaluate if physics-informed models have an advantage in predicting extreme weather is also something that we would have done with more time.

Our results indicate that the current model struggles with extreme values due to smoothing from the loss function, so experimenting with custom loss functions that prioritize extremes, as Fuxi-Extreme, would be an important avenue of work. HR-Extreme provides an opportunity to expand on Fuxi-Extreme by testing custom loss functions on a broad range of extreme weather events, rather than just heat. Exploring the tradeoff between extreme and general weather performance using finetuning and loss functions would be a natural part of this work.

After our evaluation experiments, we had aimed to extend ClimODE if there was time to address the handling of vector features. Since we did not have enough time, this would be an avenue for future work. ClimODE's architecture takes as input the concatenation of the following features: current weather variables, their gradients, velocity, and spatiotemporal embedding. The velocity is handled as multiple scalar features rather than a coupled vector quantity. Equivariant GNNs ensure that the vector features are transformed in a way that respects the underlying symmetries of the system [15, 17]. After implementing the GNN component, we could develop an equivariant version to enhance performance. Specifically, we would focus on the EGNN architecture, a prominent model that updates both scalar features and node coordinates in an invariant and equivariant manner. [15].

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A Event-Level Performance

Tables 4-7 show the performances of pre-trained and fine-tuned ClimODE across different extreme weather events in HR-Extreme.

Table 4: RMSE performance of ClimODE across weather event types in HR-Extreme.

Event	t2m	u10	v10	z	t	#
Cold	16.04	15.21	8.66	251.82	11.03	357
Debris Flow	17.12	12.28	8.74	141.43	12.34	491
Flash Flood	16.53	12.21	8.56	152.21	15.20	3197
Flood	16.40	12.38	8.61	145.82	14.05	3231
Funnel Cloud	14.30	11.79	8.27	164.30	16.32	158
Hail	12.74	10.66	8.75	165.38	15.76	1744
Heat	5.75	10.77	12.27	241.28	17.06	432
Heavy Rain	16.78	11.95	8.70	140.20	13.37	2301
Lightning	14.42	11.05	8.66	160.98	16.31	388
Marine Hail	11.31	8.93	9.63	185.05	17.67	17
Marine High Wind	14.38	11.35	8.63	179.77	13.80	89
Marine Strong Wind	24.19	11.36	11.76	163.48	19.23	11
Marine Thunderstorm Wind	16.71	12.02	9.09	148.97	13.80	994
Thunderstorm Wind	13.49	10.96	8.70	164.12	15.54	3404
Tornado	14.14	11.52	9.06	161.48	15.91	700
Waterspout	21.40	12.88	8.02	142.15	17.42	265
Wind	9.34	10.93	9.20	153.58	10.27	511

Table 5: ACC performance of ClimODE across weather event types in HR-Extreme.

Event	t2m	u10	v10	z	t	#
Cold	0.44	0.32	0.40	0.21	0.23	357
Debris Flow	0.38	0.43	0.42	0.23	0.24	491
Flash Flood	0.43	0.32	0.38	0.23	0.23	3197
Flood	0.42	0.35	0.42	0.29	0.25	3231
Funnel Cloud	0.48	0.32	0.37	0.29	0.27	158
Hail	0.48	0.37	0.41	0.31	0.29	1744
Heat	0.58	0.54	0.58	0.17	0.16	432
Heavy Rain	0.38	0.31	0.38	0.22	0.24	2301
Lightning	0.51	0.37	0.43	0.29	0.31	388
Marine Hail	0.50	0.57	0.55	0.26	0.23	17
Marine High Wind	0.17	0.36	0.40	0.28	0.26	89
Marine Strong Wind	0.49	0.22	0.45	0.30	0.34	11
Marine Thunderstorm Wind	0.37	0.37	0.37	0.20	0.19	994
Thunderstorm Wind	0.48	0.37	0.43	0.32	0.27	3404
Tornado	0.45	0.35	0.40	0.32	0.25	700
Waterspout	0.42	0.42	0.41	0.20	0.26	265
Wind	0.48	0.22	0.34	0.28	0.23	511

Table 6: RMSE performance of fine-tuned ClimODE across weather event types in HR-Extreme.

Event	t2m	u10	v10	z	t	#
Cold	16.60	13.72	5.39	433.32	12.55	357
Debris Flow	12.31	8.18	6.14	130.58	10.88	491
Flash Flood	12.13	8.10	6.10	106.88	10.28	3197
Flood	11.95	8.71	6.09	143.33	10.27	3231
Funnel Cloud	10.67	7.34	5.97	92.31	9.15	158
Hail	10.08	6.67	5.76	86.16	9.46	1744
Heat	7.98	5.25	6.66	70.81	5.72	432
Heavy Rain	12.13	8.00	5.79	114.96	10.72	2301
Lightning	10.96	7.35	6.00	91.08	10.11	388
Marine Hail	10.36	5.31	5.20	75.58	6.93	17
Marine High Wind	11.33	8.46	5.82	156.50	10.75	89
Marine Strong Wind	16.77	7.90	8.77	36.63	8.47	11
Marine Thunderstorm Wind	12.71	7.74	6.17	110.64	9.86	994
Thunderstorm Wind	10.81	6.89	5.87	88.53	9.42	3404
Tornado	11.12	7.36	6.22	95.78	9.96	700
Waterspout	14.39	9.08	6.42	117.26	9.50	265
Wind	7.72	6.22	4.99	101.34	11.35	511

Table 7: ACC performance of fine-tuned ClimODE across weather event types.

Event	t2m	u10	v10	z	t	#
Cold	0.50	0.40	0.47	0.21	0.28	357
Debris Flow	0.44	0.59	0.55	0.31	0.34	491
Flash Flood	0.45	0.46	0.48	0.24	0.29	3197
Flood	0.46	0.51	0.53	0.33	0.33	3231
Funnel Cloud	0.52	0.49	0.47	0.25	0.32	158
Hail	0.51	0.52	0.51	0.30	0.35	1744
Heat	0.66	0.66	0.64	0.20	0.27	432
Heavy Rain	0.41	0.47	0.49	0.23	0.29	2301
Lightning	0.55	0.50	0.53	0.30	0.38	388
Marine Hail	0.54	0.61	0.60	0.24	0.29	17
Marine High Wind	0.29	0.51	0.48	0.30	0.26	89
Marine Strong Wind	0.57	0.33	0.48	0.40	0.31	11
Marine Thunderstorm Wind	0.41	0.55	0.49	0.24	0.24	994
Thunderstorm Wind	0.51	0.51	0.52	0.31	0.33	3404
Tornado	0.48	0.50	0.51	0.31	0.31	700
Waterspout	0.45	0.57	0.51	0.20	0.32	265
Wind	0.52	0.41	0.45	0.27	0.30	511

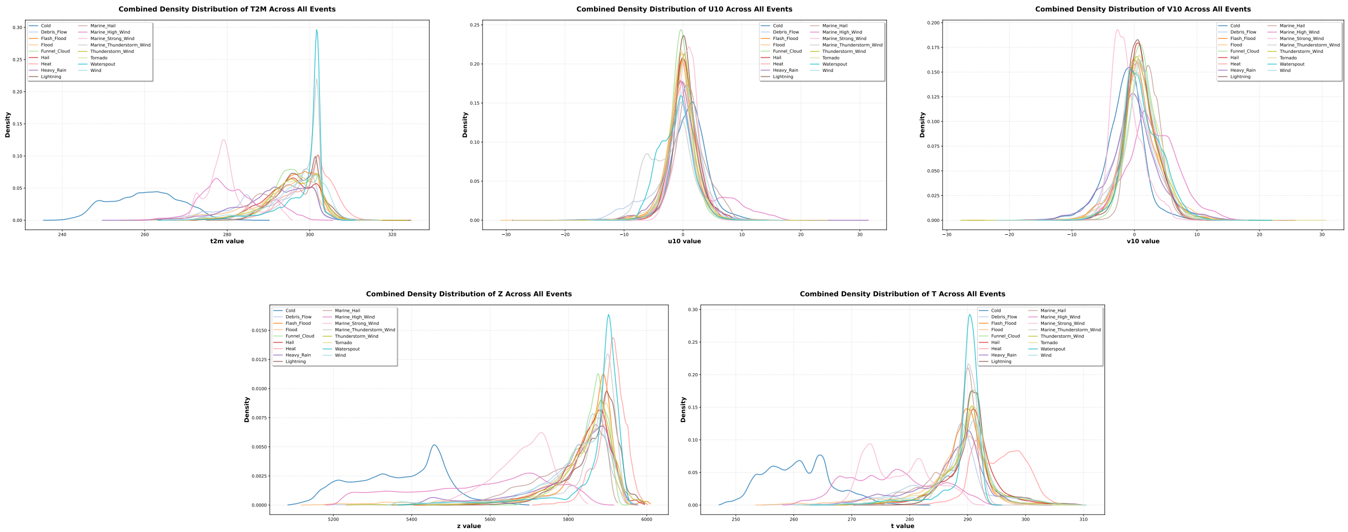

Figure 3: Distribution of each variable across each event type in HR-Extreme